

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
df = pd.read_csv("insurance.csv")
```

df

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

1338 rows × 7 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

> EDA

↳ 12 cells hidden

✓ Data Cleaning And Pre-processing 🧹🌸

```
#this data is already cleaned but lets see
```

```
df_cleaned = df.copy()
```

```
df_cleaned.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

```
#dropping duplicates
```

```
df.shape
```

(1338, 7)

```
df_cleaned.drop_duplicates(inplace = True)
```

```
df_cleaned.shape
```

```
(1337, 7)
```

```
df_cleaned.isnull().sum()
```

```
      0
age    0
sex    0
bmi    0
children 0
smoker 0
region 0
charges 0
```

```
dtype: int64
```

```
#encoding
```

```
df_cleaned['sex'].value_counts()
```

```
      count
sex
male      675
female    662
```

```
dtype: int64
```

```
df_cleaned['sex'] = df_cleaned['sex'].map({'male':0, 'female':1})
```

```
df_cleaned.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	1	27.900	0	yes	southwest	16884.92400
1	18	0	33.770	1	no	southeast	1725.55230
2	28	0	33.000	3	no	southeast	4449.46200
3	33	0	22.705	0	no	northwest	21984.47061
4	32	0	28.880	0	no	northwest	3866.85520

Next steps:

[Generate code with df_cleaned](#)[New interactive sheet](#)

```
df_cleaned['smoker'].value_counts()
```

```
      count
smoker
no      1063
yes      274
```

```
dtype: int64
```

```
df_cleaned['smoker'] = df_cleaned['smoker'].map({'yes':1, 'no':0})
```

```
df_cleaned.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	1	27.900	0	1	southwest	16884.92400
1	18	0	33.770	1	0	southeast	1725.55230
2	28	0	33.000	3	0	southeast	4449.46200
3	33	0	22.705	0	0	northwest	21984.47061
4	32	0	28.880	0	0	northwest	3866.85520

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

```
df_cleaned.rename(columns={'sex': 'is female',
                          'smoker': 'is smoker'}, inplace=True)
```

```
df_cleaned.head()
```

	age	is female	bmi	children	is smoker	region	charges	
0	19	1	27.900	0	1	southwest	16884.92400	
1	18	0	33.770	1	0	southeast	1725.55230	
2	28	0	33.000	3	0	southeast	4449.46200	
3	33	0	22.705	0	0	northwest	21984.47061	
4	32	0	28.880	0	0	northwest	3866.85520	

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

```
df['region'].value_counts()
```

	count
region	
southeast	364
southwest	325
northwest	325
northeast	324

```
dtype: int64
```

```
#here i do one hot encoding
```

```
df_cleaned = pd.get_dummies(df_cleaned, columns=['region'], drop_first=True)
```

```
df_cleaned.head()
```

	age	is female	bmi	children	is smoker	charges	region_northwest	region_southeast	region_southwest	
0	19	1	27.900	0	1	16884.92400	False	False	True	
1	18	0	33.770	1	0	1725.55230	False	True	False	
2	28	0	33.000	3	0	4449.46200	False	True	False	
3	33	0	22.705	0	0	21984.47061	True	False	False	
4	32	0	28.880	0	0	3866.85520	True	False	False	

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

```
# to turn True False into 1 and 2
df_cleaned = df_cleaned.astype(int)
```

```
#Final cleaned version
df_cleaned.head()
```

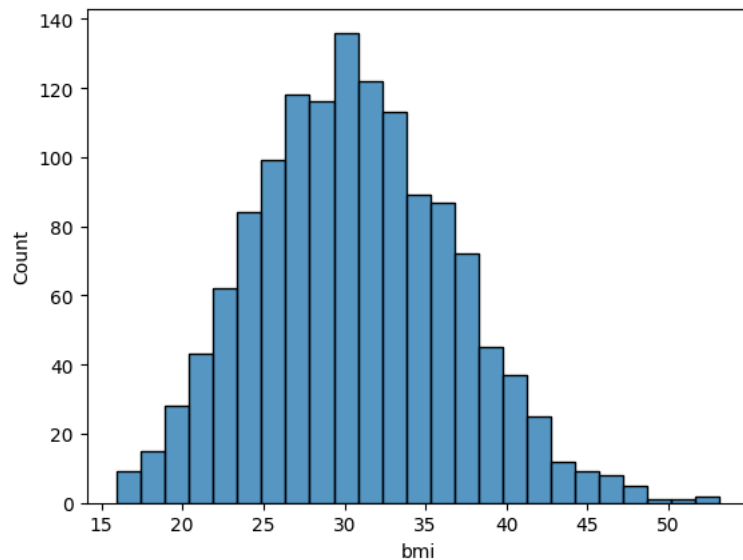
	age	is female	bmi	children	is smoker	charges	region_northwest	region_southeast	region_southwest	
0	19	1	27	0	1	16884	0	0	1	
1	18	0	33	1	0	1725	0	1	0	
2	28	0	33	3	0	4449	0	1	0	
3	33	0	22	0	0	21984	1	0	0	
4	32	0	28	0	0	3866	1	0	0	

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

Feature Engineering and Extraction ✂🔧

```
sns.histplot(df['bmi'])
```

<Axes: xlabel='bmi', ylabel='Count'>



```
#lets make bmi categories
```

```
df_cleaned['bmi_category'] = pd.cut(df_cleaned['bmi'], bins=[0, 18.5, 24.9, 29.9, np.inf], labels=['Underweight', 'Normal', 'Overweight', 'Obese'])
```

```
#used these codes to covert not showing due to some error
# df_cleaned = pd.get_dummies(df_cleaned, columns=['bmi_category'], drop_first=True)
# df_cleaned = df_cleaned.astype(int)
```

```
df_cleaned.head()
```

	age	is female	bmi	children	is smoker	charges	region_northwest	region_southeast	region_southwest	bmi_category_Normal	bmi_category_Overweight	bmi_category_Obese
0	19	1	27	0	1	16884	0	0	1	0	0	0
1	18	0	33	1	0	1725	0	1	0	0	0	0
2	28	0	33	3	0	4449	0	1	0	0	0	0
3	33	0	22	0	0	21984	1	0	0	1	0	0
4	32	0	28	0	0	3866	1	0	0	0	0	0

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

```
### Feature scaling
```

```
from sklearn.preprocessing import StandardScaler
cols = ['age', 'bmi', 'children']
scaler = StandardScaler()
df_cleaned[cols] = scaler.fit_transform(df_cleaned[cols])
```

```
df_cleaned.head()
```

	age	is female	bmi	children	is smoker	charges	region_northwest	region_southeast	region_southwest	bmi_category
0	-1.440418	1	-0.517949	-0.909234	1	16884	0	0	1	
1	-1.511647	0	0.462463	-0.079442	0	1725	0	1	0	
2	-0.799350	0	0.462463	1.580143	0	4449	0	1	0	
3	-0.443201	0	-1.334960	-0.909234	0	21984	1	0	0	
4	-0.514431	0	-0.354547	-0.909234	0	3866	1	0	0	

Next steps: [Generate code with df_cleaned](#) [New interactive sheet](#)

#you cant memorize all these just extract from Gpt Google if yk the theory
#here pearson correlation

```
from scipy.stats import pearsonr

# Pearson Correlation Calculation
# List of features to check against target
selected_features = [
    'age',
    'bmi',
    'children',
    'is female',
    'is smoker',
    'region_northwest',
    'region_southeast',
    'region_southwest',
    'bmi_category_Normal',
    'bmi_category_Overweight',
    'bmi_category_Obese'
]

# Calculate Pearson correlation for each selected feature against 'charges'
correlations = {}
for feature in selected_features:
    correlation, _ = pearsonr(df_cleaned[feature], df_cleaned['charges'])
    correlations[feature] = correlation

# Convert to DataFrame and sort in descending order
correlation_df = pd.DataFrame(list(correlations.items()), columns=['Feature', 'Pearson Correlation'])
correlation_df = correlation_df.sort_values(by='Pearson Correlation', ascending=False).reset_index(drop=True)

print("Pearson Correlation with 'charges' (Descending Order):")
display(correlation_df)
```

Pearson Correlation with 'charges' (Descending Order):

	Feature	Pearson Correlation	
0	is smoker	0.787234	
1	age	0.298309	
2	bmi_category_Obese	0.200348	
3	bmi	0.196236	
4	region_southeast	0.073577	
5	children	0.067390	
6	region_northwest	-0.038695	
7	region_southwest	-0.043637	
8	is female	-0.058046	
9	bmi_category_Normal	-0.104042	
10	bmi_category_Overweight	-0.120601	

Next steps: [Generate code with correlation_df](#) [New interactive sheet](#)

if something over +0.50 or negative affect more than -0.50
these are highly correlated features

```
# Define the list of categorical features already present in df_cleaned
cat_features = [
    'is female',
    'is smoker',
    'region_northwest',
    'region_southeast',
    'region_southwest',
    'bmi_category_Normal',
    'bmi_category_Overweight',
    'bmi_category_Obese'
]
```

	chi2_statistic	p_value	Decision
is smoker	848.219178	0.0	Reject Null (Keep Feature)
region_southeast	15.998167	0.001135	Reject Null (Keep Feature)
is female	10.258784	0.01649	Reject Null (Keep Feature)
bmi_category_Obese	8.515711	0.036473	Reject Null (Keep Feature)
region_southwest	5.091893	0.165191	Accept Null (Drop Feature)
bmi_category_Overweight	4.25149	0.235557	Accept Null (Drop Feature)
bmi_category_Normal	3.708088	0.29476	Accept Null (Drop Feature)
region_northwest	1.13424	0.768815	Accept Null (Drop Feature)

	age	bmi	children	charges	is smoker	region_southeast	is female	bmi_category_Obese	region_southwest	
0	-1.440418	-0.517949	-0.909234	16884	1	0	1	0	1	
1	-1.511647	0.462463	-0.079442	1725	0	1	0	1	0	
2	-0.799350	0.462463	1.580143	4449	0	1	0	1	0	
3	-0.443201	-1.334960	-0.909234	21984	0	0	0	0	0	
4	-0.514431	-0.354547	-0.909234	3866	0	0	0	0	0	

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